ENGINEERING APPRAISAL MEMORANDUM **5/03/2018**

To: Hannah Thomson

From: **Ronald Uy**
(Peer reviewed by Vijay Kunaseelan)

Application Number(s):	SUB60315119
Applicant:	Raymond Charles Nicklin
Proposed Activity(s):	Rural subdivision to establish 5 freehold rural-residential allotments on the basis of protecting significant native bush.
Address:	53B Remiger Road, Puhoi

As requested, I have assessed the application against Council's relevant engineering policies. These include:

- **Access**

Access to the proposed lots via the existing right of way / vehicle crossing branching off from the existing metaled Remiger Road. On-site parking.

- **Earthworks**

Earthworks for the creation / upgrade of the new ROW anticipated not to exceed the permitted activity thresholds.

- **Geotechnical Issues**

Category B soils – slightly expansive. Geotech report identified category M soils in terms of AS2870:2011. Site away from a northern allochthon overlay. Proposed building platform sites ranging from moderately sloping (1:5.5) to relatively flat. Geotechnical Investigation Report prepared by Engineering Geology Ltd., ref: 8447, dated 14 September 2017.

- **Wastewater**

On-site effluent treatment and disposal system.

- **Stormwater**

Roof runoff from dwellings on lots 1-5 will flow to reuse tanks and overflow will be piped for sheetflow discharge over pasture. Driveway / paving runoff on lots 1-5 will bypass tank system and piped to a level spreader dispersal device.

<u>Flooding and Overland Flowpaths</u>

Council Geomaps indicated overland flow paths and floodplains within the property but the proposed building platforms are away from them. The vehicle crossing and the first 70 meters (roughly) of the

ROW however are within the flood plains overlay. Site is away from coastal inundation.

- **Water Supply**

Roof rain runoff reuse tanks.

- **Services**

Power and telephone connections will be required for each lot.

- **Other matters**

As part of this assessment, a site visit to the subject site was carried out on 26 March 2018.

Having assessed the application, I recommend that, subject to the following conditions the consent can be granted from an engineering perspective.

Engineering Conditions - To be completed before issue of the s.224(c) certificate

(construction of right of way) The right of way over parts of proposed Lot 6 up to the intersection at Remiger Road shall be upgraded to a rural metaled standard to the Council's "Standards for Engineering Design and Construction". Sections with a grade greater than 1 in 8 shall be to a rural concrete standard. The upgrade shall ensure that the formed carriageway is no less than 3 metres in width. Further it shall include strategically placed passing bay/s for the purpose of servicing lots and make adequate provision for the drainage of surface water as well as provision of any ducts required for power and communications.

(engineering plans) The engineering works required by this consent shall comply with the Council's "Standards for Engineering Design and Construction" as may be amended from time to time. Engineering Plans, as specified in the "Standards", shall be submitted to the Consents Engineer, and approval thereto received in writing, prior to the commencement of any works on the site.

Any variation or changes to the approved engineering plans shall be submitted for approval as an Amendment and approval received thereto prior to construction of the varied works.
The term 'engineering works' includes, but is not limited to:
1.1 Earthworks,
1.2 Any other works required by conditions of this consent.

Note: Structures such as retaining walls, in-ground walls and bridges will require a separate Building Consent.

Note: The plans required under this condition are separate to, and do not form part of, any Building Consent that may be required on the subject site.

(silt retention) Before commencement of any works, adequate silt retention structures as detailed in Technical Publication No. 90 "Erosion and Sediment Control Guidelines for Land Disturbing Activities in the Auckland Region" shall be installed. These structures shall be maintained and cleaned out as necessary until such time as complete grass cover, or other non-erodible surface has been re-established over the area. The location and detail of the sediment control shall be included in the Engineering Plans for approval prior to commencement of construction.

(provide for electric power) Written confirmation shall be provided from the electricity network supplier responsible for the area, that provision of an electric supply has been made available by underground means to all saleable lots created and that all the network supplier's requirements for making such means of supply available have been met or satisfactory arrangements have been concluded with the Consent Holder to complete the provision of the supply.

(provide for telecommunications) Written confirmation shall be provided from the telecommunications network supplier responsible for the area, that provision of telephone services has been made available by underground means to all saleable lots created and that all the network supplier's requirements for making such services available have been met or satisfactory arrangements have been concluded with the Consent Holder to complete the provision of the service.

(new road name) The Consent Holder shall suggest to the Council a name for the new road shown on the Scheme Plan after consulting the local iwi for comment, together with clearance from Addresses@linz.govt.nz so that duplication of the name in any other part of the Auckland region is avoided. The Council Surveyor should be consulted in the first instance in regard to the appropriateness of the name and road type. When a name has been resolved with Local Board approval, the Consent Holder shall erect nameplates, in accordance with the Council's "Standards for Engineering Design and Construction".

Ongoing Conditions/Consent Notices

a) (firefighting supply) Upon the construction of habitable buildings, sufficient water volume shall be provided in accordance with NZFS Fire Fighting Water Supplies CoP SNZ 4509:2008 and that this water supply be accessible for firefighting purposes. Should the water supply be provided by way of tank storage, this storage must be located a safe distance away from any habitable dwelling in accordance with the above CoP.

b) (building restrictions) Any buildings erected on the building sites identified on the plan shall be subject to the requirements of the report prepared by Engineering Geology Ltd., ref: 8447, dated 14 September 2017, and any subsequent reports. Copies of the said plan and report(s) will be held at the offices of the Council, Centreway Road, Orewa.

c) (building restrictions - stormwater control) All stormwater from buildings and paved areas on Lots 1-5 shall be collected and disposed of in accordance with the report prepared by the Stormwater Hydrology Report prepared by Cato Bolam Consultants, ref: 40320, dated 17 October 2017, unless otherwise approved by Council. The collection and disposal system shall be installed in conjunction with the erection of any buildings and shall thereafter be maintained to the specified capacity and standard in perpetuity.

Advise Notes: